

# The exact multivariate state density (Taylor-tree Stage 1a–b)

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## Abstract

Units 223–224 open the Taylor-tree programme (Astra #26 confirmed the density-first plan). The state density of a monomial,  $v(\tau) = \int \delta(\tau - u^{2k}) u^h du$  on  $(0, 1]^d$ , is represented *exactly* as a finite list of triples  $(\mu, j, c)$  standing for  $\sum c \tau^{\mu-1} (-\log \tau)^j$  (**PowLogRep**). Peeling one coordinate of weight  $t^{w-1}$  is the multiplicative convolution  $(\text{powLogConv}_w v)(z) = \int_z^1 t^{w-1} v(z/t) dt$ , and acts on representations by an explicit linear map (**PowLogRep.conv**, **eval\_conv**): the kernel identity  $t^{w-1} (z/t)^{\mu-1} (-\log(z/t))^j = z^{\mu-1} t^{\alpha-1} (\log t - \log z)^j$ ,  $\alpha = w - \mu + 1$ , reduces everything to  $G_j(z) = \int_z^1 t^{\alpha-1} (\log t - \log z)^j dt$  with  $G_0 = (1 - z^\alpha)/\alpha$ ,  $G_{j+1} = (-\log z)^{j+1}/\alpha - \frac{j+1}{\alpha} G_j$  ( $\alpha \neq 0$ ) and  $G_j = (-\log z)^{j+1}/(j+1)$  ( $\alpha = 0$ ). Hence the new exponent  $w+1$  enters only at logarithmic degree 0 and the two invariants hold term by term: exponents are among the  $w_i + 1$  and degrees are below the multiplicity (**stateDensityRep\_exponent\_mem**, **stateDensityRep\_degree\_lt**). The measure-theoretic bridge (**weightedBoxIntegral\_eq\_stateDensity**) is the recursion of the equal-ratio product density (units 165–175) with the concrete representation in place of  $(-\log z)^{d-1}/(d-1)!$ :  $\int_{(0,1]^{n+1}} \prod a_i^{w_i} g(\prod a_i) da = \int_0^1 v(z) g(z) dz$  for every measurable  $g \geq 0$ ; through the coordinatewise power substitution this is the monomial state density with  $\mu_i = (h_i + 1)/(2k_i)$  and Jacobian  $\prod 1/(2k_i)$  (**monomialBoxIntegral\_eq\_stateDensity**). No Mellin inversion is used. Zero sorry/axiom; 9101 jobs.

## 1 The things formalised

```
def powLogBasis j := ^(-1) * (-log ) ^j; abbrev PowLogRep := List ( × × )
PowLogRep.eval, smul, shift ( ^a · ), eval_append/smul/shift/flatMap
integral_G_zero / integral_G_succ / integral_G_zero_exp -- the G_j recursion
def gRep j : PowLogRep (representation of z G_j(z)); eval_gRep; gRep_mem
def powLogConv w v z := t in Icc z 1, t^(w-1) * v (z/t)
def PowLogRep.basisConv w ( ,j,c); def PowLogRep.conv w c := c.flatMap (basisConv w)
theorem eval_basisConv
theorem PowLogRep.eval_conv : eval (conv w c) z = powLogConv w (eval c) z
theorem conv_exponent_mem : exponents insert (w+1) P
theorem conv_degree_lt : degree < r + (if = w+1 then 1 else 0)
def stateDensityRep : (n) → (Fin (n+1) → ) → PowLogRep
-- base [(w0+1,0,1)]; step conv (w 0) (rep (tail w))
theorem stateDensityRep_nonneg; PowLogRep.measurable_eval
theorem weightedBoxIntegral_eq_stateDensity :
  weightedBoxIntegral (n+1) w g
  = z in Ioc 0 1, ofReal (eval (stateDensityRep n w) z) * g z
theorem monomialBoxIntegral_eq_stateDensity (Jacobian 1/(2k), weights (h+1)/(2k) - 1)
theorem stateDensityRep_exponent_mem : t rep, i, t.1 = w i + 1
def expMult w := #{i | w i + 1 = }
theorem stateDensityRep_degree_lt : t.2.1 < expMult w t.1
```

## 2 Mathematics

A first design used the kernel  $(\log t)^i(-\log z)^{j-i}$  after a binomial expansion; that produces terms  $\tau^w(-\log \tau)^{i'}$  at the new exponent with  $i'$  up to  $j$ , which cancel in the sum but would break a term-by-term multiplicity invariant. Integrating  $(\log t - \log z)^j$  directly, the boundary term at  $t = z$  vanishes for  $j \geq 1$ , so only  $(-\log z)^j/\alpha$  survives at  $t = 1$  and the  $z^\alpha$  term appears once, at degree 0: partial fractions with a simple new pole. Numerical replica of the recursion against box quadrature: weights  $(\frac{1}{2}, \frac{1}{2})$ ,  $(-\frac{1}{2}, \frac{1}{4})$ ,  $(0, 0, 0)$ ,  $(0, 0, 1)$ ,  $(-\frac{1}{2}, 0, 1)$  agree to  $10^{-8}$ , with exponent/degree sets  $\{(1.5, 1)\}$ ,  $\{(0.5, 0), (1.25, 0)\}$ ,  $\{(1, 2)\}$ ,  $\{(1, 0), (1, 1), (2, 0)\}$ ,  $\{(0.5, 0), (1, 0), (2, 0)\}$ . Astra #26's cautions for later stages are recorded in the tide-log: phase order does not raise the exponent in several variables (truncate by spectral exponent, sum all phase orders below the cutoff); complete moments must not be summed before a spectral cutoff; holomorphy on the open polydisc gives weighted summability only at smaller radii; Mellin Laurent coefficients are density coefficients times factorials.

## 3 Lean notes

- `Real.rpow_sub ht` without explicit exponents rewrites the first matching  $t^{x-y}$  (here  $t^{w-1}$ ), and `ring` normalises real powers unexpectedly; pass the exponents and `generalize` the powers to atoms before `ring`.
- `Laplace.Grammar.powLog` already exists (`QuotientCalculus`); the basis is `powLogBasis`. A `grep -q "import ...StateDensity"` prefix-matched `StateDensityUnique` and skipped the registration (fixed in c7efae6): match the full line.
- `Nested List.maps` destructure as `obtain s, a, ha, rfl, rfl`.
- `div_eq_mul_inv` rewrites the first division pattern; pass the arguments.

## 4 Status

Stage 1a–b done. Next: integrability of the density from the box identity with  $g = 1$ , the real-valued identity for nonnegative bounded  $f$ , the Mellin transform  $\int_0^1 \tau^s v = \prod 1/(s + w_i + 1)$ , and the factorial-normalised leading coefficient (Headline XXI link); then Stage 2.